

Humic Nitisols:

Humic Nitisols are deep, well-drained tropical soils with a reddish-brown to dark reddish color due to high iron oxide content, and a clayey nitric subsurface horizon with strong, polyhedral blocky structure.

They are named “Humic” because they have relatively high organic matter content in the surface horizon, giving them darker color and higher nutrient-holding capacity compared to Dystric and Eutric Nitisols.

The deep, porous solum and stable soil structure support unrestricted deep rooting and make these soils highly productive.

Drainage:

- **Wet season:** Well-drained; water infiltrates easily due to stable structure.
- **Dry season:** Soil remains friable and workable; moderate moisture retention supports crop growth.
- **Landscape:** Typically found on undulating to hilly tropical terrain; rarely in depressions.

Depth:

- Very deep (>150 cm).
- Rooting depth is generally unrestricted due to stable, porous structure.

Workability:

- **When wet:** Plastic and sticky but manageable; does not smear like Vertisols.
- **When dry:** Firm, friable, easy to till; seedbed preparation is straightforward.

Nutrient Richness & Cation Exchange Capacity (CEC):

- CEC: Moderate to high (15–35 cmol(+)/kg).
- Base saturation: Moderate to high (>35%).
- High surface organic matter and nutrient-holding capacity - They have over a depth of 100 cm from the soil surface, more than 1.4 percent organic carbon (by weight) in the fine earth fraction
- Nitrogen and potassium levels are moderate;
- phosphorus availability may be limited due to P-fixation.

- Iron oxides dominate; micronutrients (Zn, Mn) are generally sufficient.
- Contain 4.0 percent or more 'free' iron

pH:

- Slightly acidic to near neutral (5.5–6.8).

Management:

- Highly productive soils in the humid tropics; deep, porous solum allows deep rooting and provides erosion resistance.
- Suitable for plantation crops such as cocoa, coffee, rubber, pineapple, and smallholder food crops.
- High phosphorus fixation requires careful P-fertilizer application: slow-release rock phosphate combined with smaller doses of soluble superphosphate.
- Incorporate organic matter (manure, compost, crop residues) to improve fertility, structure, and biological activity.
- Maintain vegetation cover or cover crops on slopes to reduce erosion.
- Liming may be applied if soil acidity increases beyond optimal crop levels.
- Excellent workability allows tilling within 24 hours after wetting without deterioration of structure.